PADHAKU
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CHAPTER 10

The Other Side of Zero

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
Progress is saved automatically in your browser — pick up where you left off.

⭐ More and more numbers!

? **Can there be a number less than 0? Can you think of any ways to have less than 0 of something?**

💡 **SAMPLE ANSWER**

Yes. For example, a floor below the ground floor, a temperature colder than 0°C, or owing money (having less than ₹0).

Such numbers are called **negative numbers**.

⭐ 10.1 Bela's Building of Fun

? **Some floors of the 'Building of Fun' are below the ground. What are the shops on these floors? What is there on the ground floor?**

💡 **SAMPLE ANSWER**

Below the ground: Toy Store, Video Games shop, Cinema Centre (and other floors below).

Ground floor: the **Welcome Hall** (the entry).

(check the picture in your book)

? **What do you press to go four floors up? What do you press to go three floors down?**

✔ **ANSWER**

Four floors up: + + + + = **+ 4**

Three floors down: – – – = **– 3**

? **The ground floor is called Floor 0. Can you see why?**

✔ ANSWER

To reach the ground floor from the ground floor you press no button: 0 presses.

It is the starting point (reference), so it is **Floor 0**.

? **Number all the floors in the Building of Fun.**

💡 SAMPLE ANSWER

Floor	Place
+ 5	Sports Centre
+ 3	Book Store
+ 2	Art Centre
+ 1	Food Court
0	Welcome Hall
– 1	Toy Store
– 2	Video Games shop
– 3	Cinema Centre

The building has floors – 5 to + 6. Write the names of the other floors from the picture. (check the picture in your book)

Addition to keep track of movement

? **Start from the Food Court and press + 2 in the lift. Where will you reach?**

✔ ANSWER

(+ 1) + (+ 2) = + 3

You reach Floor + 3, the **Book Store**.

Figure it Out · Addition to keep track of movement

1 **You start from Floor + 2 and press – 3 in the lift. Where will you reach? Write an expression for this movement.**

✔ ANSWER

Starting Floor + Movement = Target Floor

(+ 2) + (– 3) = – 1

You reach **Floor – 1 (Toy Store)**.

2 **Evaluate these expressions:**

a. (+ 1) + (+ 4) b. (+ 4) + (+ 1) c. (+ 4) + (– 3) d. (– 1) + (+ 2)

e. (– 1) + (+ 1) f. 0 + (+ 2) g. 0 + (– 2)

✔ ANSWER

a. **+ 5**

b. **+ 5**

c. **+ 1**

d. **+ 1**

e. **0**

f. **+ 2**

g. **– 2**

3 **Starting from different floors, find the movements required to reach Floor – 5 and write the expressions. (Example: (+ 2) + (– 7) = – 5)**

💡 **SAMPLE ANSWER**

- (+ 1) + (– 6) = – 5
- 0 + (– 5) = – 5
- (– 2) + (– 3) = – 5
- (+ 4) + (– 9) = – 5
- (+ 6) + (– 11) = – 5

🧮 Figure it Out · Combining button presses is also addition

a–d **Evaluate as the resulting movement of combining button presses:**
a. (+ 1) + (+ 4) b. (+ 4) + (+ 1)
c. (+ 4) + (– 3) + (– 2) d. (– 1) + (+ 2) + (– 3)

✔ **ANSWER**

a. **+ 5**

b. **+ 5**

c. (+ 4) + (– 3) = + 1; (+ 1) + (– 2) = **– 1**

d. (– 1) + (+ 2) = + 1; (+ 1) + (– 3) = **– 2**

⭐ Back to zero!

? **If Basant now presses + 4 and then presses − 4 in the lift, where will he reach?**

✔ ANSWER

(+ 4) + (− 4) = 0

He comes back to the **ground floor (Floor 0)**.

? **Write the inverses of these numbers: + 4, − 4, − 3, 0, + 2, − 1.**

✔ ANSWER

Number	Inverse
+ 4	− 4
− 4	+ 4
− 3	+ 3
0	0
+ 2	− 2
− 1	+ 1

? **Connect the inverses by drawing lines.**

Top: + 5, − 7, − 8, + 9

Bottom: − 9, + 8, − 5, + 7

✔ ANSWER

- **+ 5 ↔ − 5**
- **− 7 ↔ + 7**
- **− 8 ↔ + 8**

- **+ 9 ↔ − 9**

⭐ Comparing numbers using floors

? **Who is on the lowest floor?**

1. Jay is in the Art Centre, Floor + 2.
2. Asin is in the Sports Centre, Floor ___.
3. Binnu is in the Cinema Centre, Floor ___.
4. Aman is in the Toy Store, Floor ___.

✔ ANSWER

2. **+ 5**
3. **− 3**
4. **− 1**

− 3 is the smallest, so **Binnu** is on the lowest floor.

? **Should we write − 3 < − 4 or − 4 < − 3?**

✔ ANSWER

Floor − 4 is lower than Floor − 3.

So **− 4 < − 3** (also − 3 > − 4).

Figure it Out · Comparing numbers using floors

1 Fill in < or >:

a. − 2 ☐ + 5 b. − 5 ☐ + 4 c. − 5 ☐ − 3

d. + 6 ☐ − 6 e. 0 ☐ − 4 f. 0 ☐ + 4

✔ ANSWER

a. − 2 **<** + 5

b. − 5 **<** + 4

c. − 5 **<** − 3

d. + 6 **>** − 6

e. 0 **>** − 4

f. 0 **<** + 4

2 Imagine more floors. Fill in < or >:

a. − 10 ☐ − 12 b. + 17 ☐ − 10 c. 0 ☐ − 20

d. + 9 ☐ − 9 e. − 25 ☐ − 7 f. + 15 ☐ − 17

✔ ANSWER

a. − 10 **>** − 12

b. + 17 **>** − 10

c. 0 **>** − 20

d. + 9 **>** − 9

e. − 25 **<** − 7

f. + 15 **>** − 17

3 **If Floor A = – 12, Floor D = – 1 and Floor E = + 1 in the building shown as a line, find the numbers of Floors B, C, F, G and H.**

💡 **SAMPLE ANSWER**

Count the floor marks from A (– 12) upwards; D is – 1, the next mark is 0, then E is + 1.

B = – 9, C = – 6, F = + 2, G = + 6, H = + 11 (check the picture in your book)

4 **Mark the following floors of the building shown on the right: a. – 7 b. – 4 c. + 3 d. – 10**

💡 **SAMPLE ANSWER**

Count from D (– 1) or A (– 12):

a. – 7: 5 floors above A (6 floors below D)

b. – 4: 3 floors below D

c. + 3: 2 floors above E

d. – 10: 2 floors above A

⭐ Subtraction to find which button to press

? **Evaluate 15 – 5, 100 – 10 and 74 – 34 from this perspective (finding the missing number to be added).**

✔ **ANSWER**

- 5 + ? = 15, so 15 – 5 = **10**
- 10 + ? = 100, so 100 – 10 = **90**

- 34 + ? = 74, so 74 − 34 = **40**

Figure it Out · Subtraction to find which button to press

a–j Complete these expressions (Target Floor − Starting Floor = Movement needed):

a. (+ 1) − (+ 4) b. (0) − (+ 2) c. (+ 4) − (+ 1) d. (0) − (− 2)

e. (+ 4) − (− 3) f. (− 4) − (− 3) g. (− 1) − (+ 2) h. (− 2) − (− 2)

i. (− 1) − (+ 1) j. (+ 3) − (− 3)

✔ ANSWER

a. from + 4 to + 1: 3 floors down = **− 3**

b. from + 2 to 0: **− 2**

c. from + 1 to + 4: **+ 3**

d. from − 2 to 0: **+ 2**

e. from − 3 to + 4: **+ 7**

f. from − 3 to − 4: **− 1**

g. from + 2 to − 1: **− 3**

h. same floor: **0**

i. from + 1 to − 1: **− 2**

j. from − 3 to + 3: **+ 6**

Figure it Out · Adding and subtracting larger numbers

a–g Complete these expressions:

a. (+ 40) + ___ = + 200 b. (+ 40) + ___ = − 200

c. (− 50) + ___ = + 200 d. (− 50) + ___ = − 200

e. (– 200) – (– 40) f. (+ 200) – (+ 40) g. (– 200) – (+ 40)

✔ ANSWER

a. **+ 160**

b. 40 down to 0, then 200 more down: **– 240**

c. 50 up to 0, then 200 more up: **+ 250**

d. **– 150**

e. from – 40 to – 200: **– 160**

f. **+ 160**

g. from + 40 to – 200: **– 240**

⭐ Adding, subtracting, and comparing any numbers

a–h **Evaluate by imagining a suitable lift:**
a. – 125 + (– 30) b. + 105 – (– 55) c. + 105 + (+ 55) d. + 80 – (– 150)
e. + 80 + (+ 150) f. – 99 – (– 200) g. – 99 + (+ 200) h. + 1500 – (– 1500)

✔ ANSWER

a. **– 155**

b. from – 55 to + 105: 55 + 105 = **+ 160**

c. **+ 160**

d. from – 150 to + 80: 150 + 80 = **+ 230**

e. **+ 230**

f. from – 200 to – 99: **+ 101**

g. **+ 101**

h. from – 1500 to + 1500: **+ 3000**

? **In the other exercises above, did you notice that subtracting a negative number was the same as adding the corresponding positive number?**

✔ ANSWER

Yes. For example, + 105 – (– 55) = + 160 = + 105 + (+ 55), and – 99 – (– 200) = + 101 = – 99 + (+ 200).

Math Talk **Take a look at the 'infinite lift'. Does it remind you of a number line? In what ways?**

💡 SAMPLE ANSWER

Yes. It has 0 in the middle, positive numbers on one side and negative numbers on the other.

The levels are equally spaced and go on forever both ways, just like a number line (turned upright).

⭐ Back to the number line

? **If, from 5 you wish to go over to 9, how far must you travel along the number line?**

✔ ANSWER

4 steps forward: 5 + 4 = 9, and 9 – 5 = 4.

? **From 9, if you wish to go to 3, how much must you travel along the number line?**

✔ ANSWER

6 steps backward, i.e. **– 6**: 9 + (– 6) = 3, and 3 – 9 = – 6.

? **From 3, if you wish to go to – 2, how far must you travel?**

✔ ANSWER

5 steps backward, i.e. **– 5**: 3 + (– 5) = – 2, and – 2 – 3 = – 5.

Figure it Out · Back to the number line

1 **Mark 3 positive numbers and 3 negative numbers on the number line.**

💡 SAMPLE ANSWER

Positive: **2, 5, 8**

Negative: **– 1, – 3, – 7**

(You may choose other numbers.)

2 **Write the 3 marked negative numbers in the boxes: ☐ < ☐ < ☐**

💡 SAMPLE ANSWER

– 7 < – 3 < – 1

3 **Is 2 > – 3? Why? Is – 2 < 3? Why?**

✔ ANSWER

Yes, 2 > – 3, because 2 is to the right of – 3 on the number line.

Yes, – 2 < 3, because – 2 is to the left of 3.

4 **What are a. – 5 + 0 b. 7 + (– 7) c. – 10 + 20 d. 10 – 20 e. 7 – (– 7) f. – 8 – (– 10)?**

✔ ANSWER

a. **– 5**

b. **0**

c. **10**

d. **– 10**

e. 7 + 7 = **14**

f. – 8 + 10 = **2**

⭐ Using the unmarked number line to add and subtract

a–d **Use unmarked number lines to evaluate:**
a. – 125 + (– 30) b. + 105 – (– 55)
c. + 80 – (– 150) d. – 99 – (– 200)

✔ ANSWER

a. move 30 left from – 125: **– 155**

b. from – 55 to + 105: **+ 160**

c. from – 150 to + 80: **+ 230**

d. from – 200 to – 99: **+ 101**

⭐ 10.2 The Token Model

? **The lift attendant starts on Floor 0 and finds 5 positive and 3 negative tokens in his pocket. On which floor is he now?**

✔ ANSWER

3 zero pairs cancel, 2 positive tokens are left.

(+ 5) + (– 3) = + 2, so he is on **Floor + 2**.

🧮 Figure it Out · Using tokens for addition

1 **Complete the additions using tokens: a. (+ 6) + (+ 4) b. (– 3) + (– 2) c. (+ 5) + (– 7) d. (– 2) + (+ 6)**

✔ ANSWER

a. 10 positives: **+ 10**

b. 5 negatives: **– 5**

c. 5 zero pairs cancel, 2 negatives left: **– 2**

d. 2 zero pairs cancel, 4 positives left: **+ 4**

2 **Cancel the zero pairs in the two sets of tokens. On what floor is the lift attendant in each case? Write the addition statement.**

💡 SAMPLE ANSWER

a. (+ 3) + (– 5) = – 2 → **Floor – 2**

b. (+ 6) + (– 3) = + 3 → **Floor + 3**

(check the picture in your book)

⭐ Using tokens for subtraction

? **Is (– 7) – (– 5) the same as (– 7) + (+ 5)?**

✔ ANSWER

(– 7) – (– 5): take away 5 negatives from 7 negatives, 2 negatives left = – 2.

(– 7) + (+ 5): 5 zero pairs cancel, 2 negatives left = – 2.

Yes, both are – 2.

🧮 Figure it Out · Using tokens for subtraction

1 **Evaluate using tokens: a. (+ 10) – (+ 7) b. (– 8) – (– 4) c. (– 9) – (– 4) d. (+ 9) – (+ 12) e. (– 5) – (– 7) f. (– 2) – (– 6)**

✔ ANSWER

a. take 7 positives from 10: **+ 3**

b. take 4 negatives from 8: **– 4**

c. take 4 negatives from 9: **– 5**

d. add 3 zero pairs, take 12 positives: **– 3**

e. add 2 zero pairs, take 7 negatives: **+ 2**

f. add 4 zero pairs, take 6 negatives: **+ 4**

2 **Complete the subtractions: a. (– 5) – (– 7) b. (+ 10) – (+ 13) c. (– 7) – (– 9) d. (+ 3) – (+ 8) e. (– 2) – (– 7) f. (+ 3) – (+ 15)**

✔ ANSWER

a. **+ 2**

b. **– 3**

c. **+ 2**

d. **– 5**

e. **+ 5**

f. **– 12**

Figure it Out · Subtracting with zero pairs

1 **Try to subtract: – 3 – (+ 5). How many zero pairs will you have to put in? What is the result?**

✔ ANSWER

Start with 3 negatives. There are no positives, so put in **5 zero pairs**.

Take away 5 positives: 3 + 5 = 8 negatives are left.

Result: **– 8**

2 **Evaluate using tokens: a. (– 3) – (+ 10) b. (+ 8) – (– 7) c. (– 5) – (+ 9) d. (– 9) – (+ 10) e. (+ 6) – (– 4) f. (– 2) – (+ 7)**

✔ ANSWER

a. **– 13**

b. **+ 15**

c. **– 14**

d. **– 19**

e. **+ 10**

f. **– 9**

⭐ 10.3 Integers in Other Places · Credits and debits

? **Balance starts at ₹100. You deposit ₹60 (credit). Your new bank balance is ___.**

✔ ANSWER

100 + 60 = **₹160**

? **You pay an electric bill of ₹30 (debit). Your bank balance is now ___.**

✔ ANSWER

160 – 30 = **₹130**

? **You make a purchase of ₹150 (debit). What is your bank balance now? Is this possible?**

✔ ANSWER

130 – 150 = **– ₹20**

Yes, some banks allow the balance to become negative for a short time (they may charge a fee or interest).

? **Next day you make ₹200 at your business. What is your balance now?**

✔ ANSWER

– 20 + 200 = **₹180**

Figure it Out · Credits and debits

1 **Start with ₹0. Credits of ₹30, ₹40, ₹50 and debits of ₹40, ₹50, ₹60. What is your balance now?**

✔ ANSWER

Credits: 30 + 40 + 50 = + 120

Debits: 40 + 50 + 60 = 150, i.e. – 150

(+ 120) + (– 150) = **– ₹30**

2 **Start with ₹0. Debits of ₹1, 2, 4, 8, 16, 32, 64 and 128, then a single credit of ₹256. What is your balance now?**

✔ ANSWER

Debits: 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 = 255, i.e. – 255

(– 255) + (+ 256) = **+ ₹1**

3 **Why is it generally better to maintain a positive balance in your bank account? When may it be worthwhile to temporarily have a negative balance?**

💡 **SAMPLE ANSWER**

A negative balance means you owe the bank; banks may charge interest or a fee, so you lose money.

A positive balance keeps your money safe for needs and emergencies.

A short negative balance may be useful for an urgent need (like a medical emergency) or a business purchase that will soon bring in more money, as in the example above.

🧮 Figure it Out · Geographical cross sections

1 **Looking at the geographical cross section, fill in the heights of A, B, C, D, E, F, G.**

💡 **SAMPLE ANSWER**

Approximate heights (read from the scale):

Point	Height
A	+ 1500 m
B	– 500 m
C	+ 300 m
D	– 1200 m
E	+ 1200 m
F	– 200 m

G	+ 100 m

(check the picture in your book)

2 **Which is the highest point in this geographical cross section? Which is the lowest point?**

✔ ANSWER

Highest point: **A**

Lowest point: **D**

3 **Write the points A, B, ..., G in decreasing order of heights, and in increasing order.**

💡 SAMPLE ANSWER

Decreasing: **A, E, C, G, F, B, D**

Increasing: **D, B, F, G, C, E, A**

4 **What is the highest point above sea level on Earth? What is its height?**

✔ ANSWER

Mount Everest (Himalayas).

Height: about **+ 8849 m** above sea level (older books: 8848 m).

5 **What is the lowest point with respect to sea level on land or on the ocean floor? What is its height?**

✔ ANSWER

Ocean floor: **Challenger Deep** in the Mariana Trench (Pacific Ocean), about **– 10,994 m** (nearly 11 km below sea level).

On land: the shore of the Dead Sea, about – 430 m.

⭐ Temperature

? **What do you think will be the temperature during the summer when you feel very hot? What has been the maximum temperature in summer and the minimum in winter last year in your area? Find out.**

💡 SAMPLE ANSWER

When it feels very hot, the temperature is often about 40°C or more.

Example (Delhi): maximum about 45°C in summer, minimum about 4°C in winter. Check the news or a weather app for your own area.

🧮 Figure it Out · Temperature

1 **Find out the places in India where temperatures sometimes go below 0°C. What**

is common among these places? Why does it become colder there and not in other places?

💡 SAMPLE ANSWER

Places: Leh, Kargil and Dras (Ladakh), Gulmarg and Srinagar (Kashmir), Manali and Shimla (Himachal Pradesh), Tawang (Arunachal Pradesh), North Sikkim.

Common: they are high up in the Himalayan mountains in the north.

The air is colder at greater heights, and these places get snow in winter, so the temperature can fall below 0°C.

2 **Match the temperatures in Leh on a November day with the time:**

Temperatures: 14°C, 8°C, – 2°C, – 4°C

Times: 02:00 a.m., 11:00 p.m., 02:00 p.m., 11:00 a.m.

✔ ANSWER

Temperature	Time
14°C	02:00 p.m. (warmest, afternoon)
8°C	11:00 a.m.
– 2°C	11:00 p.m.
– 4°C	02:00 a.m. (coldest, late night)

10.4 Explorations with Integers · A hollow integer grid

? Second grid: top row 5, − 3, − 5; middle row 0, − 5; bottom row − 8, − 2, 7. Fill in:

Top row: 5 + (− 3) + (− 5) = ___

Bottom row: (− 8) + (− 2) + 7 = ___

Left column: 5 + 0 + (− 8) = ___

Right column: (− 5) + (− 5) + 7 = ___

✔ ANSWER

Top row: **− 3**

Bottom row: **− 3**

Left column: **− 3**

Right column: **− 3**

Figure it Out · A hollow integer grid

1 Do the calculations for the second grid above and find the border sum.

✔ ANSWER

Each row and column adds to − 3.

Border sum = **− 3**

2 Complete the grids to make the required border sum:

Grid 1 (sum + 4): − 10 at top-left, − 5 at middle-right, 9 at bottom-left

Grid 2 (sum − 2): 6 and 8 in the top row, − 5 at middle-right, − 2 at bottom-middle

Grid 3 (sum − 4): 7 in the top row, − 5 at middle-right

💡 SAMPLE ANSWER

One way for each (rows written top to bottom):

Grid 1:

− 10	10	4
5		− 5
9	− 10	5

Check: − 10 + 10 + 4 = 4; 9 − 10 + 5 = 4; − 10 + 5 + 9 = 4; 4 − 5 + 5 = 4

Grid 2:

6	8	− 16
11		− 5
− 19	− 2	19

Check: 6 + 8 − 16 = − 2; − 19 − 2 + 19 = − 2; 6 + 11 − 19 = − 2; − 16 − 5 + 19 = − 2

Grid 3:

7	− 2	− 9
− 3		− 5
− 8	− 6	10

Check: 7 − 2 − 9 = − 4; − 8 − 6 + 10 = − 4; 7 − 3 − 8 = − 4; − 9 − 5 + 10 = − 4

(check the picture in your book for where the given numbers are)

3 **For the last grid above, find more than one way of filling the numbers to get border sum – 4.**

💡 **SAMPLE ANSWER**

Way 2:

7	– 1	– 10
– 1		– 5
– 10	– 5	11

Way 3:

7	0	– 11
0		– 5
– 11	– 5	12

Every row and column adds to **– 4**.

4 **Which other grids can be filled in multiple ways? What could be the reason?**

✔ **ANSWER**

All the grids can be filled in many ways.

Reason: there are more empty boxes than conditions (only 4 sums must be equal), so we can choose some numbers freely and then find the rest.

5 **Make a border integer square puzzle and challenge your classmates.**

💡 **SAMPLE ANSWER**

Example: border sum − 6. Give 2 at top-left, − 4 at middle-right and 0 at bottom-middle; ask friends to fill the rest.

One answer: top row 2, − 3, − 5; middle 1, − 4; bottom − 9, 0, 3.

Figure it Out · An amazing grid of numbers!

1 **Try afresh, choose different numbers this time. What sum did you get? Was it different from the first time? Try a few more times!**

💡 **SAMPLE ANSWER**

Example: circle 3, − 1, − 2, − 1 (one from each row and column): 3 + (− 1) + (− 2) + (− 1) = − 1.

The sum is **always − 1**, the same as the first time.

2 **Play the same game with the grids below. What answer did you get?**

Grid A: 7, 10, 13, 16 / − 2, 1, 4, 7 / − 11, − 8, − 5, − 2 / − 20, − 7, − 14, − 11

Grid B: − 11, − 10, − 9, − 8 / − 7, − 6, − 5, − 4 / − 3, − 2, − 1, 0 / 1, 2, 3, 4

💡 **SAMPLE ANSWER**

Grid A: e.g. 7 + 1 + (– 5) + (– 11) = **– 8**

Grid B: e.g. (– 11) + (– 6) + (– 1) + 4 = **– 14**

Each grid always gives the same sum. (In Grid A the second number of the last row fits the pattern only as – 17; check it in your book.)

3 **What could be so special about these grids? Is the magic in the numbers or the way they are arranged or both? Can you make more such grids?**

💡 **SAMPLE ANSWER**

It is the arrangement: every row goes up by the same steps, and each row is the row above plus (or minus) a fixed number.

So any 4 numbers taken one from each row and each column always add up to the same total.

To make one: write 4 numbers along the top and 4 down the side; each box = its top number + its side number. Example: top 0, 1, 2, 3 and side 0, – 5, 4, – 2 gives rows 0 1 2 3 / – 5 – 4 – 3 – 2 / 4 5 6 7 / – 2 – 1 0 1 (sum always – 3 + 6 = **3**).

Figure it Out · Integers (end exercises)

1 **Write all the integers between the given pairs, in increasing order: a. 0 and – 7 b. – 4 and 4 c. – 8 and – 15 d. – 30 and – 23**

✔ ANSWER

a. **– 6, – 5, – 4, – 3, – 2, – 1**

b. **– 3, – 2, – 1, 0, 1, 2, 3**

c. **– 14, – 13, – 12, – 11, – 10, – 9**

d. **– 29, – 28, – 27, – 26, – 25, – 24**

2 **Give three numbers such that their sum is – 8.**

💡 SAMPLE ANSWER

(– 5) + (– 2) + (– 1) = – 8

Another: (– 10) + 7 + (– 5) = – 8

3 **Two dice have faces – 1, 2, – 3, 4, – 5, 6. Smallest sum – 10, largest 12. Which numbers between – 10 and + 12 are not possible as sums?**

✔ ANSWER

Possible sums:

- with – 5: – 10, – 8, – 6, – 3, – 1, 1
- with – 3: – 6, – 4, – 1, 1, 3
- with – 1: – 2, 1, 3, 5
- with 2: 4, 6, 8
- with 4: 8, 10

- 6 + 6 = 12

Not possible: **– 9, – 7, – 5, 0, 2, 7, 9, 11**

4 **Solve these:**

8 – 13 (– 8) – (13) (– 13) – (– 8) (– 13) + (– 8)

8 + (– 13) (– 8) – (– 13) (13) – 8 13 – (– 8)

✔ **ANSWER**

8 – 13 = **– 5**

(– 8) – 13 = **– 21**

(– 13) – (– 8) = – 13 + 8 = **– 5**

(– 13) + (– 8) = **– 21**

8 + (– 13) = **– 5**

(– 8) – (– 13) = – 8 + 13 = **5**

13 – 8 = **5**

13 – (– 8) = 13 + 8 = **21**

5 **Find the years below. (Hint: there was no year 0.)**

a. From the present year, which year was it 150 years ago?

b. From the present year, which year was it 2200 years ago?

c. What will be the year 320 years after 680 BCE?

✔ **ANSWER**

Taking the present year as 2026 CE:

a. 2026 – 150 = **1876 CE**

b. 2026 – 2200 = – 174. Since there is no year 0, go one more year back: **175 BCE**

c. 680 – 320 = 360, still before CE: **360 BCE**

(Use your own present year for a and b.)

6 **Complete the sequences:**

a. (– 40), (– 34), (– 28), (– 22), ___, ___, ___

b. 3, 4, 2, 5, 1, 6, 0, 7, ___, ___, ___

c. ___, ___, 12, 6, 1, (– 3), (– 6), ___, ___, ___

✔ ANSWER

a. add 6 each time: **– 16, – 10, – 4**

b. two patterns (3, 2, 1, 0, ... and 4, 5, 6, 7, ...): **– 1, 8, – 2**

c. differences are – 6, – 5, – 4, – 3, so before 12 they were – 8, – 7 and next are – 2, – 1, 0: **27, 19**, 12, 6, 1, – 3, – 6, **– 8, – 9, – 9**

7 **Six integer cards: (+ 1), (+ 7), (+ 18), (– 5), (– 2), (– 9). Pick cards and make an expression using addition(s) and subtraction(s) whose value is close to (– 30).**

💡 SAMPLE ANSWER

(– 2) + (– 9) – (+ 18) – (+ 1) = – 11 – 18 – 1 = **– 30** (exactly – 30)

8 **(positive integer) – (positive integer) can be positive or negative. What about: a. (positive)**

– (negative) b. (positive) + (negative) c. (negative) + (negative) d. (negative) – (negative) e. (negative) – (positive) f. (negative) + (positive)

✔ ANSWER

a. **Always positive** (e.g. 3 – (– 2) = 5)

b. **Positive, negative or zero** (5 + (– 2) = 3; 2 + (– 5) = – 3)

c. **Always negative** (– 3 + (– 2) = – 5)

d. **Positive, negative or zero** (– 2 – (– 5) = 3; – 5 – (– 2) = – 3)

e. **Always negative** (– 3 – 2 = – 5)

f. **Positive, negative or zero** (– 2 + 5 = 3; – 5 + 2 = – 3)

9 **This string has a total of 100 tokens arranged in a particular pattern. What is the value of the string?**

💡 SAMPLE ANSWER

Cancel the zero pairs in each repeating group and multiply by the number of groups.

Value of the string = **+ 20** (check the picture in your book)

Figure it Out · Brahmagupta's Rules

1 **Can you explain each of Brahmagupta's rules in terms of Bela's Building of Fun, or in terms**

of a number line?

💡 **SAMPLE ANSWER**

Addition:

- Positive + positive: going up, then up more: you end above 0.
- Negative + negative: going down, then down more: you end below 0.
- Positive + negative: up and down cancel floor by floor; you move in the direction of the bigger number by the difference.
- Number + its inverse: going up and coming down the same number of floors brings you back to 0.
- Number + 0: pressing no button keeps you on the same floor.

Subtraction (Target – Start = Movement):

- Larger – smaller positive: you must go up, so positive. Smaller – larger: you must go down, so negative.
- Subtracting a negative: e.g. 2 – (– 3) means going from – 3 up to 2, i.e. 3 up to 0 then 2 more = 2 + 3.
- Number – itself: start and target are the same floor, so 0 movement.
- Number – 0: going from 0 to the number is the number itself; 0 – number: going from the number back to 0 is its inverse.

2 **Give your own examples of each rule.**

💡 **SAMPLE ANSWER**

Addition: 4 + 6 = 10; (– 4) + (– 6) = – 10; (– 7) + 3 = – 4 and 7 + (– 3) = 4; 9 + (– 9) = 0; – 6 + 0 = – 6

Subtraction: 8 – 5 = 3; 5 – 8 = – 3; 4 – (– 6) = 4 + 6 = 10; – 7 – (– 7) = 0; – 9 – 0 = – 9 and 0 – (– 9) = 9

🎉 **Chapter 10 done! Revise the answers once more tomorrow.**